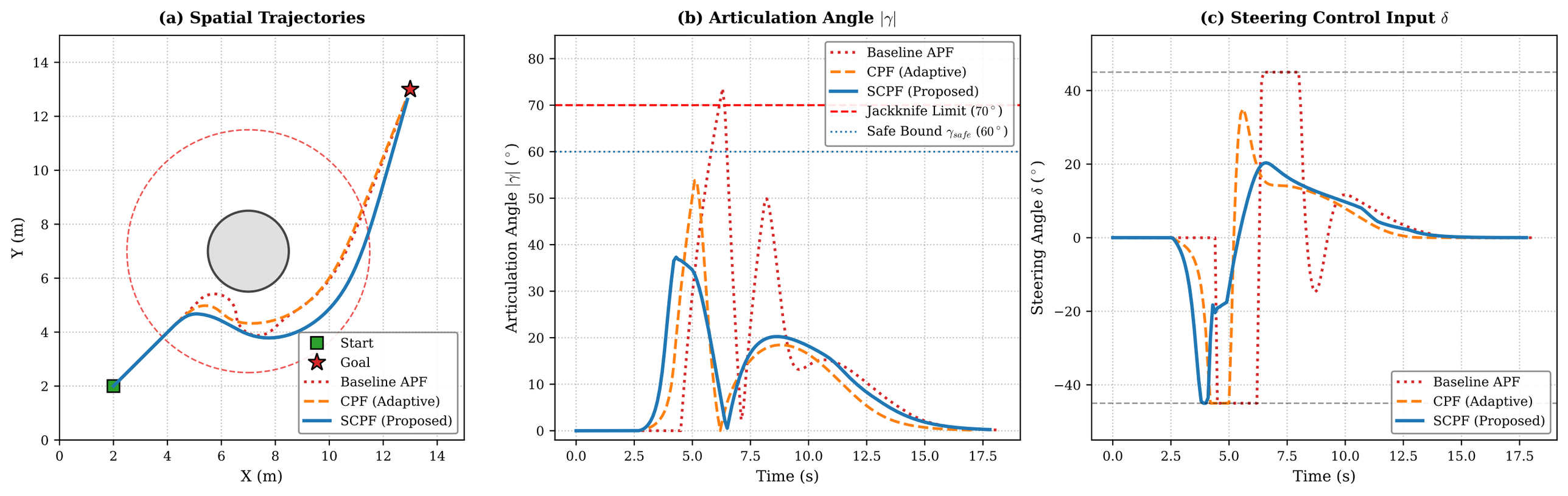
(a) Spatial Trajectories
X (m)
Y (m)
Start
Goal
Baseline APF
CPF (Adaptive)
SCPF (Proposed)
(b) Articulation Angle $|\gamma|$
Articulation Angle $|\gamma|$ (°)
Time (s)
Baseline APF
CPF (Adaptive)
SCPF (Proposed)
Jackknife Limit (70°)
Safe Bound γ_{safe} (60°)
(c) Steering Control Input δ
Steering Angle δ (°)
Time (s)
Baseline APF
CPF (Adaptive)
SCPF (Proposed)